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12430

12 - STD	QUARTERLY EXAMINATION - 2025	
TIME :3.00 Hrs	BIOLOGY	MARKS : 70

BIO - BOTANY (Marks - 35)

PART - I

I. Choose the correct answer for the following Questions. **8 x 1 = 8**

1. Parthenocarpic fruits lack
   (a) Endocarp (b) Epicarp (c) mesocarp (d) Seed.
2. Double fertilization and Triple fusion found in
   (a) Gymnosperm (b) Angiosperm (c) Pteridophyte (d) Algae
3. "Gametes are never hybrid" This is a statement of
   (a) Law of dominance (b) Law of Independent assortment
   (c) Law of Segregation (d) Law of random fertilization.
4. **Assertion (A)**: Gamma rays are generally used to induce mutation in wheat varieties. **Reason (R)** : Because they carry lower energy to non-ionize electrons from atom.
   (a) A and R are Correct. (b) A is correct. R is wrong
   (c) A is wrong. R is Correct. (d) A and R are wrong
5. Match the following

Column A		Column B
1. Exonuclease	-	(a) add or remove phosphate
2. Endonuclease	-	(b) binding the DNA fragments
3. Alkaline phosphatase	-	(c) cut the DNA at terminus
4. Ligase	-	(d) cut the DNA at middle.

   (A) 1-c, 2-d, 3-a, 4-b (B) 1-a, 2-b, 3-c, 4-d
   (C) 1-c, 2-d, 3-b, 4-a (d) 1-a, 2-c, 3-b, 4-d
6. The bacteria responsible for inducing tumours in Several dicot plants are
   (a) Candida utilis (b) Spirulina (c) Chlorella (d) Agrobacterium
7. Virus free plants are developed from
   (a) Organ Culture (b) Meristem culture
   (c) Protoplast culture (d) Cell suspension culture
8. Fusion of chloroplast is done by the Chemical.
   (a) DDT (b) EDTA (c) PEG (d) Colchicine

PART - II

II. Answer any four questions **4 x 2 = 8**

9. What is mellitophylly?
10. Give the names of the scientist who rediscovered mendelism.
11. write the stages in the mechanism of crossingover in sequence.
12. What is zymology?
13. What is Totipotency?
14. Name the chemicals used in gene transfer.

PART - III

III. Answer any Three questions. Question no-19 is compulsory. **3 x 3 = 9**

15. Draw and label the parts of an Ovule.
16. what is Back cross? mention its types.
17. Write the uses of genetic map.
18. Write the advantages of herbicide tolerant crops.
19. mention the types of Tissue culture based on Explant.

PART - IV

IV. Answer in Detail **2 X 5 = 10**

20. (a) Explain about Parthenocarpy. write its Significance **(OR)**
    (b) Explain the Dominant Epistasis with an example.
21. (a) Mention the name of man-made Cereal. How it is formed? **(OR)**
    (b) Write the Applications of plant tissue culture.

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XII - BIO - ZOOLOGY (Marks - 35)

I. Choose the Correct Answer : 8 x 1 = 8

1. Valine an aminoacid is coded by known codon _____ but first deciphered codon was _____
   A) AUU-UUC B) UUA - UUG C) AUG - UAA D) GUU - UUU
2. Paedogenetic Parthinogenesis found in _____
   A) Redialarva and larva of gallfly B) Annelids
   C) Sea urchin D) Honeybees
3. Match it

A) Vesiculase	-	1) Connects Ovary to Pelvic wall
B) Mesovarium	-	2) Sperm lysin
C) Scrotum	-	3) Enhance sperm motility
D) Hyaluronidase	-	4) Thermoregulator

   A) A-2, B-1, C-3, D-4 B) A-3, B-1, C-4, D-2
   C) A-3, B-4, C-2, D-1 D) A-1, B-3, C-4, D-2
4. **Assertion (A)** - Tubectomy in the surgical sterilisation in Human
   Reason (B) - Tubectomy prevent the entry of sperm in to urethra
   A) Both 'A' and 'R' are True and 'R' is the correct explanation of 'A'
   B) Both 'A' and 'R' are the True R is not the correct explanation of 'A'
   C) 'A' is True but 'R' is false. D) 'A' is False but 'R' is True.
5. _____ Chemical capable of stop cell division at metaphase stage.
   A) NAG - N- Acetyl galactosamine B) Phenylalanine hydroxylase.
   C) Tyrosinase D) Colchicine
6. Sudden appearance of vestigial organ in highly evolved organisms is called
   A) Atavistic Organs B) Analogous Organs
   C) Homologous Organ D) Origin of connecting link organs
7. Exaggerated response of the immune system to certain common antigen is called _____
   A) Immuno deficiency disease B) Allergy
   C) Auto immune disease D) Immuno suppressed disease
8. Cyclosporin - 'A' is an immuno suppressive drug produced from _____
   A) Aspergillus niger B) Monascus purpureus
   C) Penicillium notatum D) Trichoderma Polysporum

II. Answer any Four Questions 4 x 2 = 8

9. What is A sexual reproduction? When compared to sexual reproduction it is not given importance why?
10. What is the composition of semen in Human.
11. Male partner of a childless couple found producing semen with low sperm count but they want their own children without donor's help. What could be your suggestion.
12. Among healthy human male and female who is heterogametic in nature State reason.
13. Genetic code is universal - State reason.
14. Mention the types of Antigens with ex.

III. Answer any Three Questions. Answering Q.No.17 is compulsory. 3 x 3= 9

15. Define Bioremediation? Mention the contribution of genetically modified Pseudomonas putida and Ideonella sakiensis to it.
16. List out common withdraw symptoms of drug and alcohol abuse.
17. Why the human genome project is called a mega project?
18. One of your fellow student Jaya having treated for poisonous snake bite at Jaay clinic and recovered healthy guess what kind of immunity he might have got in hospital. Write about nature of that immunity.
19. Write short note on Lippes loop and saheli

IV. Answer in Detail 2 x 5 = 10

20. a) Explain the various phases of the menstrual cycle. **(OR)**
    b) Define aneuploidy? Write short note on Autosomal aneuploidy in human beings.
21. a) Write about modern synthetic theory of Evolution. List out the factors responsible for Evolution as mentioned in it. **(OR)**
    b) Explain the structure of an Immunoglobulin with neat labelled diagramme.

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